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Shri Vile Parle Kelavani Mandal's
DWARKADAS J. SANGHVI COLLEGE OF ENGINEERING
(Autonomous College Affiliated to the University of Mumbai)
NAAC Accredited with "A" Grade (CGPA : 3.18)
Department of Computer Engineering

Title : Quality Analysis of Borewell Water

Ishaan Sheth
Jainam Gala
Kavya Shah
Parth Mehta
Guide: Dr. Kranti Ghag

Introduction

Borewell groundwater is a critical source of fresh water, especially in rural and semi-urban India where millions depend on it for domestic, agricultural, and industrial use. However, contamination from industrial effluents, agricultural runoff, natural geological factors, and human activity poses serious health and environmental risks.

Traditional water quality testing methods - including **laboratory analysis and geophysical surveys** - are slow, expensive, labor-intensive, and incapable of continuous monitoring.

This project proposes a novel, real-time borewell water quality monitoring system combining Internet of Things (IoT) sensor networks, Artificial Intelligence, and Machine Learning to monitor critical physicochemical parameters including:

- **pH and Turbidity**
- **Dissolved Oxygen (DO)**
- **Total Dissolved Solids (TDS)**
- **Electrical Conductivity (EC)**
- **Ammonia, Hardness, and Sulphate**

The system supports Sustainable Development Goal 6 (Clean Water and Sanitation).

Contribution / Novelty

Real-Time IoT Monitoring

Continuous sensing using pH, Turbidity, TDS+EC, Dissolved Oxygen, and MQ-135 sensors connected to ESP32-S3 microcontroller with Wi-Fi transmission via MQTT and HTTP/HTTPS protocols.

AI/ML Predictive Analytics

Four forecasting models evaluated - ARIMA, SARIMA, Prophet, and LSTM - for Water Quality Index (WQI) prediction. LSTM selected with lowest RMSE (12.59), MAE (10.41), and MAPE (7.19%).

Water Quality Index (WQI)

Computed using Weighted Arithmetic Index Method across 8 parameters: DO, pH, Conductivity, Turbidity, Ammonia-N, Hardness, Sulphate, TDS.

Mobile and Web Dashboard

Real-time data visualization and threshold-based push alerts sent to users via mobile app and web portal.

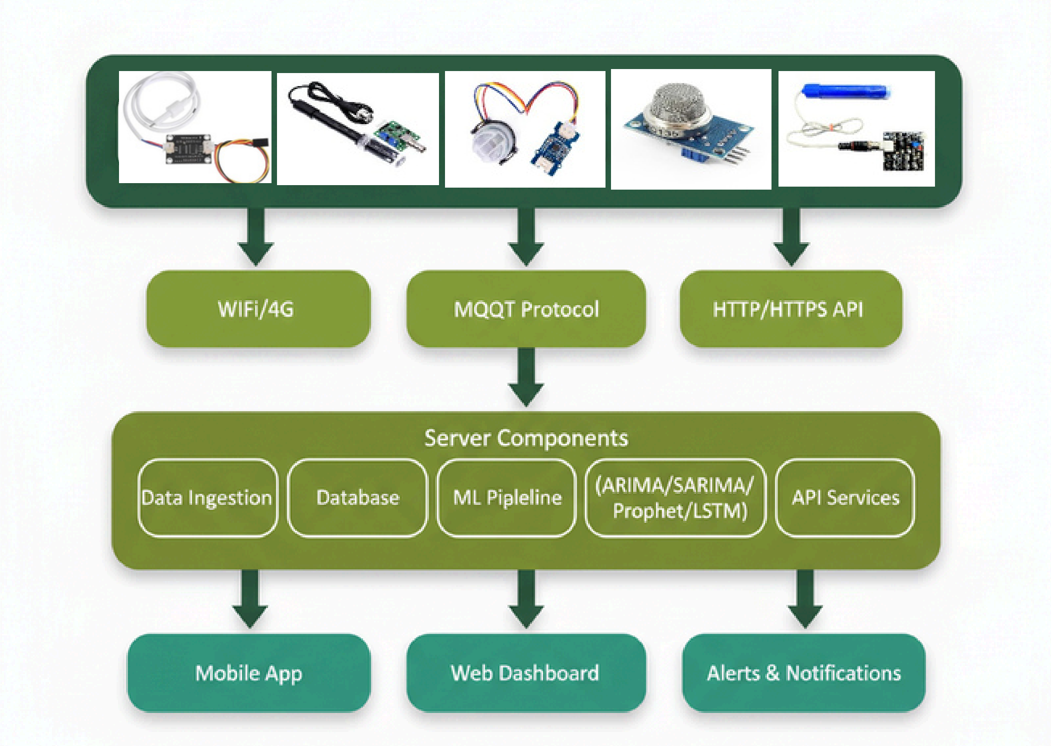
Cost-Effective and Scalable

Low-power, modular design deployable in rural and urban settings; energy-efficient with non-invasive assessment.

Dataset

Telangana State Open Data Portal (water quality data 2018-2024), pre-processed and extended for WQI modeling.

System Architecture (Block Diagram)



Results

WQI Historical Trend (2018-2024)

WQI ranged between **110-126.5** with a general rising trend indicating declining water quality. **WQI above 100** = Unsuitable for drinking without treatment.

Model Performance Comparison

Model	RMSE	MAE	MAPE	Avg CI Width
ARIMA	19.00	14.99	9.79	66.29
LSTM	12.59	10.41	7.19%	5.40
SARIMA	13.63	11.31	7.66	38.11
Prophet	16.95	14.98	10.04	22.50

LSTM achieved the best accuracy with the narrowest confidence intervals, making it the chosen model for WQI forecasting. Predicted WQI for Dec 2023-Nov 2024 ranged 131-142, closely matching actual values.

Sensor Outputs Validated

- TDS+EC Sensor: readings from **0-8 ppm** (clean water range confirmed)
- Turbidity Sensor: analog voltage output ranging **283-3652 mV**
- MQ-135 Ammonia Sensor: raw values with delta-from-baseline readings confirmed gas detection

Conclusion

This project demonstrates a complete, automated, real-time borewell water quality monitoring system that integrates IoT sensing, cloud connectivity, and AI/ML-driven predictive analytics.

The system successfully overcomes major limitations of traditional testing - high cost, time delays, and lack of continuous monitoring - by enabling instant alerts, anomaly detection, and contamination forecasting.

The LSTM model proved most reliable for WQI forecasting with the lowest error metrics and narrowest confidence intervals.

Borewell samples in the study region consistently showed WQI greater than 100, classifying water as "Unsuitable for Drinking" without treatment - confirming the urgent need for such monitoring systems.

This solution supports SDG 6 (Clean Water and Sanitation), benefits farmers, households, and industries, and is scalable for deployment across rural and urban India.

Future work will integrate:

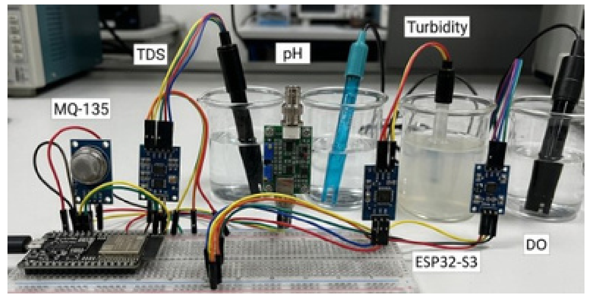
- GIS mapping for geographic visualization
- Solar power for energy efficiency
- Enhanced sensor arrays
- Broader geographic validation studies

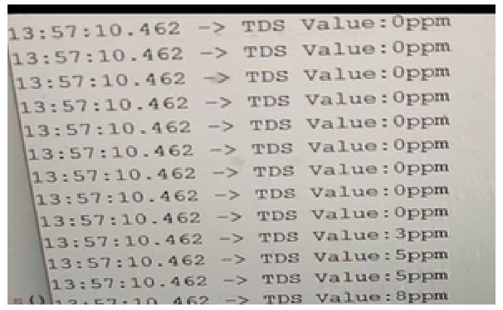
GitHub / Code Repository

Python
ESP32
TensorFlow
MQTT

GitHub Repository

Scan for Code Repository

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13:57:10.462 -> TDS Value:0ppm
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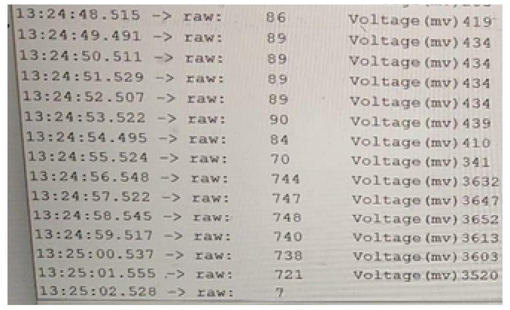
13:57:10.462 -> TDS Value:5ppm
13:57:10.462 -> TDS Value:5ppm
13:57:10.462 -> TDS Value:8ppm
```

Output of the TDS and EC Sensor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13:24:48.515 -> raw:    86    Voltage(mv)419
13:24:49.491 -> raw:    89    Voltage(mv)434
13:24:50.511 -> raw:    89    Voltage(mv)434
13:24:51.529 -> raw:    89    Voltage(mv)434
13:24:52.507 -> raw:    89    Voltage(mv)434
13:24:53.522 -> raw:    90    Voltage(mv)439
13:24:54.495 -> raw:    84    Voltage(mv)410
13:24:55.524 -> raw:    70    Voltage(mv)341
13:24:56.548 -> raw:    744   Voltage(mv)3632
13:24:57.522 -> raw:    747   Voltage(mv)3647
13:24:58.545 -> raw:    748   Voltage(mv)3652
13:24:59.517 -> raw:    740   Voltage(mv)3613
13:25:00.537 -> raw:    738   Voltage(mv)3603
13:25:01.555 -> raw:    721   Voltage(mv)3520
13:25:02.528 -> raw:    7
```

Output of the Turbidity Sensor